Chat Server

SERVER_SIDE

```
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>
#include<netinet/in.h>
#include<sys/select.h>
#include<string.h>
#define MAX 10
int main()
{
	int server_sock,client[MAX],maxfd,n;
	char buffer[1024];
	struct sockaddr_in server;
	fd_set readfds;
	server_sock=socket(AF_INET,SOCK_STREAM,0);
	server.sin_family=AF_INET;
	server.sin_port=htons(4533);
	server.sin_addr.s_addr=INADDR_ANY;
	bind(server_sock,(struct sockaddr*)&server,sizeof(server));
	listen(server_sock,10);
	printf("\nServer started at port 4533.\n");
	//Initialise
	for(int i=0;i<MAX;i++)
		client[i]=-1;
	while(1)
	{
		FD_ZERO(&readfds);
```

```
FD_SET(server_sock,&readfds);
maxfd=server_sock;
for(int i=0;i<MAX;i++)
{
        if(client[i]!=-1)
        {
                FD_SET(client[i],&readfds);
                if(client[i]>maxfd)
                maxfd=client[i];
        }
}
select(maxfd+1,&readfds,NULL,NULL,NULL);
if(FD_ISSET(server_sock,&readfds))
{
        int newclient=accept(server_sock,NULL,NULL);
        for(int i=0;i<MAX;i++)
        {
                if(client[i]==-1)
                {
                        client[i]=newclient;
                        printf("Client %d is connected\n",i+1);
                        break;
                }
         }
}
for(int i=0;i<MAX;i++)
{
        if(client[i]!=-1 && FD_ISSET(client[i],&readfds))
```

```
                {
                        int n=recv(client[i],buffer,sizeof(buffer)-1,0);
                        if(n<=0)
                        {
                                printf("Client %d disconnected..\n",i+1);
                                close(client[i]);
                                client[i]=-1;
                        }
                        else
                        {
                                buffer[n]='\0';
                                printf("Client %d : %s\n",i+1,buffer);
                                if(strcmp(buffer,"quit")==0)
                                {
                                        close(client[i]);
                                        client[i]=-1;
                                        continue;
                                }
                                printf("Server : ");
                                fgets(buffer,sizeof(buffer),stdin);
                                send(client[i],buffer,strlen(buffer),0);
                        }
                }
        }
    }
    close(server_sock);
    return 0;
}
```

CLIENT PROGRAM

```
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>
#include<netinet/in.h>
#include<string.h>
int main()
{
        int client_sock;
        struct sockaddr_in server;
        char buffer[1024];
        client_sock=socket(AF_INET,SOCK_STREAM,0);
        server.sin_family=AF_INET;
        server.sin_port=htons(4533);
        server.sin_addr.s_addr=INADDR_ANY;
        int con=connect(client_sock,(struct sockaddr*)&server,sizeof(server));
        printf("\nClient is connected..\n");
        while(1)
        {
                printf("You : ");
                fgets(buffer,sizeof(buffer),stdin);
                buffer[strcspn(buffer,"\n")]='\0';
                if(strcmp(buffer,"quit")==0)
                {
                        send(client_sock,buffer,strlen(buffer),0);
                        printf("Closing the connection..\n");
                        break;
                }
```

```
                send(client_sock,buffer,strlen(buffer),0);
                int n=recv(client_sock,buffer,sizeof(buffer)-1,0);
                buffer[n]='\0';
                printf("Server : %s\n",buffer);
        }
        return 0;
}
```

SERVER OUTPUT

```
[2024cse333@localhost cn]$ ./chat_s
Server started at port 4533.
Client 1 is connected
Client 2 is connected
Client 3 is connected
Client 3 : HI
Server : Hello 3
Client 1 : Hello server
Server : Hello dear 1
Client 2 : Hai server master
Server : hello client 2
Client 1 : quit
Client 3 : hai hdsgfk sjkd
Server : jhdsjhfsld
Client 2 : bvsjacgaj
Server : knzbckj
Client 2 : nbzscjk
Server : zj,xckj
Client 3 : quit
Client 2 : quit
```

CLIENT OUTPUT

CLIENT 1:

[2024cse333@localhost cn]$ ./chat_c

Client is connected..

You : Hello server

Server : Hello dear 1

You : quit

Closing the connection..

CLIENT 2:

[2024cse333@localhost cn]$ ./chat_c

Client is connected..

You : Hai server master

Server : hello client 2

You : bvsjacgaj

Server : knzbckj

You : nbzscjk

Server : zj,xckj

You : quit

Closing the connection..

CLIENT 3:

[2024cse333@localhost cn]$ ./chat_c

Client is connected..

You : HI

Server : Hello 3

You : hai hdsgfk sjkd

Server : jhdsjhfsld

You : quit

Closing the connection.

TIMESERVER

Server side

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
#include<unistd.h>
#include<sys/wait.h>
#include<sys/socket.h>
#include<netinet/in.h>
#include<time.h>
#include<sys/poll.h>
#define PORT 8433
#define BUFFER_SIZE 1024
int main()
{
        int server_sock;
        struct sockaddr_in server_addr,client_addr;
        char buffer[BUFFER_SIZE];
        server_sock=socket(AF_INET,SOCK_DGRAM,0);
        server_addr.sin_family=AF_INET;
        server_addr.sin_port=htons(PORT);
        server_addr.sin_addr.s_addr=INADDR_ANY;
        bind(server_sock,(struct sockaddr*)&server_addr,sizeof(server_addr));
        printf("\nUDP Time server running on PORT %d\n",PORT);
        struct pollfd fds[1];
        fds[0].fd=server_sock;
        fds[0].events=POLL_IN;
        while(1)
```

```
        {
                int activity=poll(fds,1,-1);
                if(activity<0)
                {
                        perror("Poll failed");
                        exit(0);
                }
                if(fds[0].revents & POLL_IN)
                {
                        int client_len=sizeof(client_addr);
                        memset(buffer,0,BUFFER_SIZE);
                        int n=recvfrom(server_sock,buffer,strlen(buffer),0,(struct
                        sockaddr*)&client_addr,&client_len);
                        printf("\nRecieved from client\n");
                        time_t current_time;
                        time(&current_time);
                        char * time_str=ctime(&current_time);
                        printf("%s",time_str);
                        sendto(server_sock,time_str,strlen(time_str),0,(struct
                        sockaddr*)&client_addr,client_len);
                        printf("System time sent\n");
                }
        }
        close(server_sock);
        return 0;
}
```

CLIENT SIDE

```
#include<stdio.h>
```

```
#include<stdlib.h>
#include<unistd.h>
#include<netinet/in.h>
#include<sys/wait.h>
#include<string.h>
#define PORT 8433
#define BUFFER_SIZE 1024
int main()
{
        int client_sock;
        struct sockaddr_in server_addr;
        char buffer[BUFFER_SIZE];
        client_sock=socket(AF_INET,SOCK_DGRAM,0);
        server_addr.sin_family=AF_INET;
        server_addr.sin_port=htons(PORT);
        server_addr.sin_addr.s_addr=INADDR_ANY;
        char request[]="TIME REQUEST";
        sendto(client_sock,request,sizeof(request),0,(struct
        sockaddr*)&server_addr,sizeof(server_addr));
        printf("\nRequest sent from client\n");
        int server_len=sizeof(server_addr);
        int n=recvfrom(client_sock,buffer,sizeof(buffer),0,(struct sockaddr
        *)&server_addr,&server_len);
        buffer[n]='\0';
        printf("Current system time from server\n%s\n",buffer);
        close(client_sock);
        return 0;
}
```

OUTPUT

Server Side-

UDP Time server running on PORT 8433

Recieved from client : TIME REQUEST

Mon Aug 31 12:14:47 2026

System time sent

Client side-

[2024cse333@localhost cn]$ ./time_c

Request sent from client

Current system time from server

Mon Aug 31 09:50:57 2026

FILESERVER

Server side-

```
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>
#include<string.h>
#include<netinet/in.h>
#include<sys/wait.h>
#define PORT 8499
#define BUFFER_SIZE 1024
int main()
{
	int server_sock,client_sock;
	struct sockaddr_in server_addr,client_addr;
	char filename[BUFFER_SIZE],buffer[BUFFER_SIZE];
	server_sock=socket(AF_INET,SOCK_STREAM,0);
	server_addr.sin_family=AF_INET;
	server_addr.sin_port=htons(PORT);
	server_addr.sin_addr.s_addr=INADDR_ANY;
	bind(server_sock,(struct sockaddr*)&server_addr,sizeof(server_addr));
	listen(server_sock,10);
	printf("File server running on PORT %d\n",PORT);
	while(1)
	{
		int client_len=sizeof(client_addr);
		client_sock=accept(server_sock,(struct sockaddr*)&client_addr,&client_len);
		pid_t pid=fork();
		if(pid<0)
```

```
{
        perror("Fork failed\n");
        exit(0);
}
else if(pid==0)
{
        close(server_sock);
        memset(filename,0,BUFFER_SIZE);
        int n=recv(client_sock,filename,sizeof(filename),0);
        if(n<0)
                close(client_sock);
        filename[n]='\0';
        request\n",getpid(),filename);
        FILE *fp=fopen(filename,"r");
        sprintf(buffer,"Server Process ID: %d\n",getpid());
        send(client_sock,buffer,strlen(buffer),0);
        if(fp==NULL)
        {
                sprintf(buffer,"File %s does not exists",filename);
                send(client_sock,buffer,strlen(buffer),0);
        }
        else
        {
                sprintf(buffer,"------File Contents---------\n");
                send(client_sock,buffer,strlen(buffer),0);
                while(fgets(buffer,BUFFER_SIZE-1,fp))
                {
                        send(client_sock,buffer,strlen(buffer),0);
```

```
                        }
                        fclose(fp);
                }
                close(client_sock);
                exit(0);
        }
        else
        {
                close(client_sock);
                waitpid(-1,NULL,WNOHANG);
        }
    }
    close(server_sock);
    return 0;
}
```

Client Side-

```
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>
#include<string.h>
#include<netinet/in.h>
#include<sys/wait.h>
#define PORT 8499
#define BUFFER_SIZE 1024
int main()
{
    int client_sock;
    struct sockaddr_in server_addr;
```

```
        char buffer[BUFFER_SIZE],filename[BUFFER_SIZE];
        client_sock=socket(AF_INET,SOCK_STREAM,0);
        server_addr.sin_family=AF_INET;
        server_addr.sin_port=htons(PORT);
        server_addr.sin_addr.s_addr=INADDR_ANY;
        connect(client_sock,(struct sockaddr*)&server_addr,sizeof(server_addr));
        printf("Enter the filename:");
        scanf("%s",filename);
        send(client_sock,filename,strlen(filename),0);
        printf("Filename is sent\n");
        recv(client_sock,buffer,strlen(buffer),0);
        printf("%s",buffer);
        int n;
        n=recv(client_sock,buffer,BUFFER_SIZE,0);
        buffer[n]='\0';
        printf("%s\n",buffer);
        close(client_sock);
        return 0;
}
```

OUTPUT

Server Side-

File server running on PORT 8499

Client Side-

Enter the filename:test_f.txt

Filename is sent

Server Process ID: 3799080

------File Contents---------

Hello!

Good Morning.

This is the test file for concurrent file server experiment in cn lab.

Thank you for verifying using this file.